



VII Semester MIDTERM TEST
Machine Learning (CSE_4032)

Time Duration: 2 Hours

Date: 15/9/2025

Max marks: 30 MARKS

Question No	Topic	Marks	BL	CO
1	Which of the following best describes the difference in error signals between supervised and reinforcement learning? A) Supervised learning relies on sparse, delayed error signals, while RL uses immediate and detailed labels. B) Both rely on the same gradient-based error propagation from labeled targets. C) Supervised learning uses explicit error from labels, while RL uses implicit feedback in the form of rewards or penalties. D) RL agents can compute exact loss values like cross-entropy after every action.	1	Understand	CO. 1
2	Suppose two models are compared: Model A minimizes MSE, Model B minimizes MAE. On a dataset with heavy outliers, which of the following is true? A) Model A's parameters will shift more toward fitting the outliers. B) Model B's parameters will overfit the outliers more strongly. C) Both models will ignore the outliers equally. D) Model A will approximate the median, while Model B approximates the mean.	1	Analysis	CO. 1
3	In terms of achieving generalization, which statement best explains the bias–variance tradeoff? A) Reducing bias always reduces variance. B) Increasing variance always improves generalization. C) A balance between bias and variance minimizes expected test error. D) Bias and variance are unrelated to test performance.	1	Analysis	CO. 1
4	When comparing discrimination and regression approaches to classification, which is true? A) Regression always produces linear decision boundaries, while discriminant analysis can produce nonlinear ones. B) Discriminant functions explicitly model class boundaries, while regression functions approximate target values and thresholds are imposed afterward. C) Regression requires probability density assumptions, while discriminant functions do not. D) Both are equivalent whenever the classes are balanced.	1	Analysis	CO. 2
5	Which statement best explains the relationship between gradient descent and the cost function?	1	Understand	CO. 2

	A) Gradient descent updates parameters in the direction of the gradient to maximize the cost function. B) Gradient descent uses the negative gradient of the cost function to iteratively reduce its value. C) Gradient descent moves orthogonally to the gradient to avoid saddle points. D) Gradient descent guarantees reaching the global minimum of any cost function.			
6	Suppose logistic regression is used with $h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$ If $h_{\theta}(x) = 0.9$ then the log-odds are approximately: A) 0 B) 1 C) 2.2 D) 9	1	Apply	CO. 2
7	The residual error in linear regression is defined as: $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$ This quantity is minimized across all samples by: A) Maximizing XX^T. B) Adjusting weights randomly until convergence. C) Replacing regression with classification. D) Minimizing the residual sum-of-squares criterion.	1	Analysis	CO. 2
8	For discriminant function $g(x) = -x_1 + 2x_2 - 5,$ determine the side of the boundary for point $(x_1 = 4, x_2 = 3)$. A) $g(x) = 1$, Class 1 B) $g(x) = -3$, Class 2 C) $g(x) = 0$, On boundary D) $g(x) = 7$, Class 1 side	1	Apply	CO. 3
9	The Lagrangian for the hard-margin SVM introduces multipliers. $\lambda_i \geq 0.$ Their analytical role is: A) To weight the contribution of misclassified points. B) To ensure $w^T w$ remains convex. C) To regularize against overfitting. D) To enforce linear separability constraints during optimization.	1	Understand	CO. 3
10	Consider the hyperplane $3x_1 - 4x_2 + 12 = 0$ The distance of the origin from this hyperplane is: A) 12/5 B) 3/5	1	Apply	CO. 3

	C) 4/5 D) 7/5													
11 A	Consider two datasets: I. one that is perfectly linearly separable, and II. another where classes overlap due to noise. Analyze which classifier (the maximal margin classifier or the soft margin classifier) would be more suitable for each dataset. Explain the geometric reasoning behind the choice (margin width, feasibility, robustness). Discuss the role of slack variables and the parameter CCC in the case of overlapping classes. Illustrate with clear diagrams showing the separating hyperplane, margin boundaries, and support vectors.	4	Analysis	CO. 3										
Ans.	<table><tr><th>Topic Covered</th><th>Marks</th></tr><tr><td>Correct choice for dataset (i) - States that the maximal margin (hard margin) classifier is appropriate for perfectly separable data. - Mentions widest possible margin. </td><td>0.5</td></tr><tr><td>Geometric reasoning for (i) - Explains how maximizing the margin improves robustness. - Identifies support vectors as the defining points. - Diagram shows margin boundaries and hyperplane. </td><td>0.5 0.5 0.5</td></tr><tr><td>Correct choice for dataset (ii) - States that the soft margin classifier is more appropriate for overlapping/noisy data. </td><td>0.5</td></tr><tr><td>Explanation of slack variables and C - Analyzes how slack variables allow controlled violations. - Explains role of C in balancing margin width vs errors. - Diagram shows some violators (inside or beyond margin). </td><td>0.5 0.5 0.5</td></tr></table>				Topic Covered	Marks	Correct choice for dataset (i) - States that the maximal margin (hard margin) classifier is appropriate for perfectly separable data. - Mentions widest possible margin.	0.5	Geometric reasoning for (i) - Explains how maximizing the margin improves robustness. - Identifies support vectors as the defining points. - Diagram shows margin boundaries and hyperplane.	0.5 0.5 0.5	Correct choice for dataset (ii) States that the soft margin classifier is more appropriate for overlapping/noisy data.	0.5	Explanation of slack variables and C - Analyzes how slack variables allow controlled violations. - Explains role of C in balancing margin width vs errors. - Diagram shows some violators (inside or beyond margin).	0.5 0.5 0.5
Topic Covered	Marks													
Correct choice for dataset (i) - States that the maximal margin (hard margin) classifier is appropriate for perfectly separable data. - Mentions widest possible margin.	0.5													
Geometric reasoning for (i) - Explains how maximizing the margin improves robustness. - Identifies support vectors as the defining points. - Diagram shows margin boundaries and hyperplane.	0.5 0.5 0.5													
Correct choice for dataset (ii) States that the soft margin classifier is more appropriate for overlapping/noisy data.	0.5													
Explanation of slack variables and C - Analyzes how slack variables allow controlled violations. - Explains role of C in balancing margin width vs errors. - Diagram shows some violators (inside or beyond margin).	0.5 0.5 0.5													
11 B	An underwater monitoring system analyzes 800 hydrophone clips to detect illegal blast fishing. 12.5% of the clips contain a blast. - The system predicts “Blast” for 18% of the clips. - Of these predicted positives, 72 are correct detections. Tasks : - Using the given information, construct the confusion. - Apply the formulas for Precision, Sensitivity (True Positive Rate), and Specificity (True Negative Rate) to this scenario. - Compute the F1 score using the obtained values	3	Apply	CO. 1										
Ans.	<table><tr><th>Topic Covered</th><th>Marks</th></tr><tr><td>Correct confusion matrix construction</td><td>0.25* 4= 1</td></tr><tr><td>Precision</td><td>0.5</td></tr><tr><td>Sensitivity</td><td>0.5</td></tr></table>				Topic Covered	Marks	Correct confusion matrix construction	0.25* 4= 1	Precision	0.5	Sensitivity	0.5		
Topic Covered	Marks													
Correct confusion matrix construction	0.25* 4= 1													
Precision	0.5													
Sensitivity	0.5													

Specificity	0.5
F1 Score	0.5

Construct the confusion matrix

Here's how these values were derived:

- **Total Clips:** 800
- **Actual Positives (Clips with blasts):** 12.5% X 800 =100
- **Actual Negatives (Clips without blasts):** 800-100 =700
- **Predicted Positives:** 18% X 800 =144
- **True Positives (TP):** 72 (given)
- **False Positives (FP):** 144-72= 72
- **False Negatives (FN):** 100 – 72 = 28
- **True Negatives (TN):** 700 – 72 = 628

	Predicted Blast	Predicted Non-Blast	Total
Actual Blast	TP = 72	FN = 28	100
Actual Non-Blast	FP = 72	TN = 628	700
Total	144	656	800

- Precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{72}{72 + 72} = \frac{72}{144} = 0.50$$

- Sensitivity (Recall / True Positive Rate)

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{72}{72 + 28} = \frac{72}{100} = 0.72$$

- Specificity (True Negative Rate)

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{628}{628 + 72} = \frac{628}{700} \approx 0.897$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot 0.50 \cdot 0.72}{0.50 + 0.72} = \frac{0.72}{1.22} \approx 0.590$$

Metric	Value
Precision	0.50
Sensitivity	0.72
Specificity	0.897
F1 score	0.590

11 C	Analyze how Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO) differ in their sources of biological inspiration	3	Analyze	CO. 1
------	---	---	---------	-------

	and the way collective behavior is modeled. Highlight at least one similarity and one difference in their mechanisms.																											
	<table><tr><th>Topic Covered</th><th>Marks</th></tr><tr><td>Inspiration Sources - ACO → inspired by foraging behavior of ants (pheromone trails). - PSO → inspired by social behavior / knowledge sharing (human societies, flocking). </td><td>0.5 0.5</td></tr><tr><td>Mechanism of Collective Behavior (1 mark) - ACO → indirect communication. - PSO → direct interaction. </td><td>0.5 0.5</td></tr><tr><td>Analytical Comparison (1 mark) - One similarity (both population-based, nature-inspired, emergent collective intelligence). - One difference (indirect vs direct interaction, discrete paths vs continuous movement). </td><td>0.5 0.5</td></tr></table>	Topic Covered	Marks	Inspiration Sources - ACO → inspired by foraging behavior of ants (pheromone trails). - PSO → inspired by social behavior / knowledge sharing (human societies, flocking).	0.5 0.5	Mechanism of Collective Behavior (1 mark) - ACO → indirect communication. - PSO → direct interaction.	0.5 0.5	Analytical Comparison (1 mark) - One similarity (both population-based, nature-inspired, emergent collective intelligence). - One difference (indirect vs direct interaction, discrete paths vs continuous movement).	0.5 0.5																			
Topic Covered	Marks																											
Inspiration Sources - ACO → inspired by foraging behavior of ants (pheromone trails). - PSO → inspired by social behavior / knowledge sharing (human societies, flocking).	0.5 0.5																											
Mechanism of Collective Behavior (1 mark) - ACO → indirect communication. - PSO → direct interaction.	0.5 0.5																											
Analytical Comparison (1 mark) - One similarity (both population-based, nature-inspired, emergent collective intelligence). - One difference (indirect vs direct interaction, discrete paths vs continuous movement).	0.5 0.5																											
12 A	You are given the following training dataset for predicting house categories: <table><tr><th>ID</th><th>x</th><th>y</th><th>Category</th></tr><tr><td>A</td><td>7</td><td>8</td><td>High</td></tr><tr><td>B</td><td>6</td><td>6</td><td>Medium</td></tr><tr><td>C</td><td>5</td><td>9</td><td>Medium</td></tr><tr><td>D</td><td>9</td><td>8</td><td>High</td></tr><tr><td>E</td><td>3</td><td>4</td><td>Low</td></tr></table> Two test points are to be classified: $T_1 = (5,7)$ $T_2 = (7,6)$ Apply the k-NN classification algorithm to compute the predicted category for the two test points, T_1 and T_2 with Manhattan distance and $k=3$.	ID	x	y	Category	A	7	8	High	B	6	6	Medium	C	5	9	Medium	D	9	8	High	E	3	4	Low	5	Apply	CO. 2
ID	x	y	Category																									
A	7	8	High																									
B	6	6	Medium																									
C	5	9	Medium																									
D	9	8	High																									
E	3	4	Low																									
Ans.	<u>Classify $T_1 = (5,7)$</u> Compute distances: - To A(7,8): $5 - 7 + 7 - 8 = 2 + 1 = 3$ - To B(6,6): $5 - 6 + 7 - 6 = 1 + 1 = 2$ - To C(5,9): $5 - 5 + 7 - 9 = 0 + 2 = 2$ - To D(9,8): $5 - 9 + 7 - 8 = 4 + 1 = 5$ - To E(3,4): $5 - 3 + 7 - 4 = 2 + 3 = 5$ Distances (sorted): - B: 2 (Medium) - C: 2 (Medium) - A: 3 (High) - D: 5 (High) - E: 5 (Low)																											

k=3 nearest neighbors: B (Medium), C (Medium), A (High).

Classify T2 = (7,6)

Compute distances:

- To A(7,8): $|7 - 7| + |6 - 8| = 0 + 2 = 2$
- To B(6,6): $|7 - 6| + |6 - 6| = 1 + 0 = 1$
- To C(5,9): $|7 - 5| + |6 - 9| = 2 + 3 = 5$
- To D(9,8): $|7 - 9| + |6 - 8| = 2 + 2 = 4$
- To E(3,4): $|7 - 3| + |6 - 4| = 4 + 2 = 6$

Distances (sorted):

- B: 1 (Medium)
- A: 2 (High)
- D: 4 (High)
- C: 5 (Medium)
- E: 6 (Low)

k=3 nearest neighbors: B (Medium), A (High), D (High).

T1 (5,7) → Medium

T2 (7,6) → High

Topic Covered	Marks
Distance calculations to classify T ₁ = (5,7) - Atleast 4 correct	0.5*4=2
T ₁ classified as Medium	0.5
Distance calculations to classify T ₂ = (7,6) - Atleast 4 correct	0.5 *4 =2
T ₂ classified as High	0.5

If Euclidean used instead of Manhattan: Max Marks= 2.5

If T₁ considered to classify T₂: Max Marks= 4

12 B

Apply the law of total probability in a Bayesian Belief Network with nodes ServerOverload, ErrorLogSpike, and PagerDutyAlert. ServerOverload influences whether an ErrorLogSpike occurs. ErrorLogSpike in turn influences whether a PagerDutyAlert is triggered. The network parameters are:

- The probability of ServerOverload is 0.3.
- The probability of ErrorLogSpike given ServerOverload is 0.75.
- The probability of ErrorLogSpike given no ServerOverload is 0.15.
- The probability of PagerDutyAlert given ErrorLogSpike is 0.80.
- The probability of PagerDutyAlert given no ErrorLogSpike is 0.10.

Draw the Bayesian Belief Network diagram clearly showing the nodes and directed edges. Construct the Conditional Probability

3

Apply

CO. 2

	Tables (CPTs) for ServerOverload , ErrorLogSpike and PagerDutyAlert. Compute the probability of PagerDutyAlert given ServerOverload.																																							
Ans.	ServerOverload (SO) —► ErrorLogSpike (ELS) —► PagerDutyAlert (PDA) <table border="1"><tr><td>SO</td><td>P(SO)</td></tr><tr><td>True</td><td>0.30</td></tr><tr><td>False</td><td>0.70</td></tr></table> <table border="1"><tr><td>SO</td><td>ELS</td><td>P(ELS SO)</td></tr><tr><td>T</td><td>T</td><td>0.75</td></tr><tr><td>T</td><td>F</td><td>0.25</td></tr><tr><td>F</td><td>T</td><td>0.15</td></tr><tr><td>F</td><td>F</td><td>0.85</td></tr></table> <table border="1"><tr><td>ELS</td><td>PDA</td><td>P(PDA ELS)</td></tr><tr><td>T</td><td>T</td><td>0.80</td></tr><tr><td>T</td><td>F</td><td>0.20</td></tr><tr><td>F</td><td>T</td><td>0.10</td></tr><tr><td>F</td><td>F</td><td>0.90</td></tr></table> To compute the probability of a PagerDutyAlert given a ServerOverload, we apply the law of total probability. Since PagerDutyAlert is only directly influenced by ErrorLogSpike, we sum over the possible states of ErrorLogSpike: $P(\text{PDA} \text{SO}) = P(\text{PDA} \text{ELS}, \text{SO}) \cdot P(\text{ELS} \text{SO}) + P(\text{PDA} \neg\text{ELS}, \text{SO}) \cdot P(\neg\text{ELS} \text{SO})$ Due to the structure of the BBN, PagerDutyAlert is conditionally independent of ServerOverload given ErrorLogSpike. This simplifies the expression: $P(\text{PDA} \text{ELS}, \text{SO}) = P(\text{PDA} \text{ELS})$ $P(\text{PDA} \neg\text{ELS}, \text{SO}) = P(\text{PDA} \neg\text{ELS})$ Substituting these into the equation gives: $P(\text{PDA} \text{SO}) = P(\text{PDA} \text{ELS}) \cdot P(\text{ELS} \text{SO}) + P(\text{PDA} \neg\text{ELS}) \cdot P(\neg\text{ELS} \text{SO})$ Now, we use the values from the CPTs: • $P(\text{PDA} \text{ELS}) = 0.80$ • $P(\text{ELS} \text{SO}) = 0.75$ • $P(\text{PDA} \neg\text{ELS}) = 0.10$ • $P(\neg\text{ELS} \text{SO}) = 1 - P(\text{ELS} \text{SO}) = 1 - 0.75 = 0.25$				SO	P(SO)	True	0.30	False	0.70	SO	ELS	P(ELS SO)	T	T	0.75	T	F	0.25	F	T	0.15	F	F	0.85	ELS	PDA	P(PDA ELS)	T	T	0.80	T	F	0.20	F	T	0.10	F	F	0.90
SO	P(SO)																																							
True	0.30																																							
False	0.70																																							
SO	ELS	P(ELS SO)																																						
T	T	0.75																																						
T	F	0.25																																						
F	T	0.15																																						
F	F	0.85																																						
ELS	PDA	P(PDA ELS)																																						
T	T	0.80																																						
T	F	0.20																																						
F	T	0.10																																						
F	F	0.90																																						

	Plugging in the values: $P(\text{PDA} \text{SO}) = (0.80)(0.75) + (0.10)(0.25)$ $P(\text{PDA} \text{SO}) = 0.60 + 0.025$ $P(\text{PDA} \text{SO}) = 0.625$													
	<table><tr><th>Topic Covered</th><th>Marks</th></tr><tr><td>Correct DAG with nodes : ServerOverload → ErrorLogSpike → PagerDutyAlert</td><td>0.5</td></tr><tr><td>Correct CPTs for: ServerOverload ErrorLogSpike PagerDutyAlert</td><td>0.5 0.5 0.5</td></tr><tr><td>Correct substitution</td><td>0.5</td></tr><tr><td>Correct answer</td><td>0.5</td></tr></table>				Topic Covered	Marks	Correct DAG with nodes : ServerOverload → ErrorLogSpike → PagerDutyAlert	0.5	Correct CPTs for: ServerOverload ErrorLogSpike PagerDutyAlert	0.5 0.5 0.5	Correct substitution	0.5	Correct answer	0.5
Topic Covered	Marks													
Correct DAG with nodes : ServerOverload → ErrorLogSpike → PagerDutyAlert	0.5													
Correct CPTs for: ServerOverload ErrorLogSpike PagerDutyAlert	0.5 0.5 0.5													
Correct substitution	0.5													
Correct answer	0.5													
12 C	Evaluate the importance of the learning rate in the Perceptron update rule. What could happen if it is set too high or too low?	2	Evaluate	CO. 3										
Ans.	<table><tr><th>Topic Covered</th><th>Marks</th></tr><tr><td>Importance of Learning Rate - Learning rate controls the size of weight updates during training. - Ensures stable and gradual convergence. </td><td>0.5</td></tr><tr><td>If Too High - May cause oscillations or divergence. - Model fails to converge to a decision boundary. </td><td>0.5</td></tr><tr><td>If Too Low (0.5 mark) - Training becomes very slow. </td><td>0.5</td></tr><tr><td>Balanced Evaluation - Explains that a properly chosen learning rate ensures faster yet stable convergence. - May also mention the need for tuning/experimentation. </td><td>0.5</td></tr></table>				Topic Covered	Marks	Importance of Learning Rate - Learning rate controls the size of weight updates during training. - Ensures stable and gradual convergence.	0.5	If Too High - May cause oscillations or divergence. - Model fails to converge to a decision boundary.	0.5	If Too Low (0.5 mark) Training becomes very slow.	0.5	Balanced Evaluation - Explains that a properly chosen learning rate ensures faster yet stable convergence. - May also mention the need for tuning/experimentation.	0.5
Topic Covered	Marks													
Importance of Learning Rate - Learning rate controls the size of weight updates during training. - Ensures stable and gradual convergence.	0.5													
If Too High - May cause oscillations or divergence. - Model fails to converge to a decision boundary.	0.5													
If Too Low (0.5 mark) Training becomes very slow.	0.5													
Balanced Evaluation - Explains that a properly chosen learning rate ensures faster yet stable convergence. - May also mention the need for tuning/experimentation.	0.5													

Note: BL refers to Bloom's Taxonomy Level.